

MANAUS PONTA PELADA, Brazil

WMO: 823320

Lat: 3.146S

Lon: 59.986W

Elev: 267

StdP: 14.55

Time Zone: -4.00 (W04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	72.1	73.1	66.4	98.2	84.6	68.1	104.2	83.7	15.0	80.1	13.2	81.7	4.7	60	0.323

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	14.2	95.0	78.4	93.5	78.4	92.6	78.4	81.0	88.9	80.4	88.4	79.9	87.7	7.2	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.1	152.4	84.7	78.6	150.1	84.4	77.6	144.9	83.4	44.9	89.3	44.3	89.1	43.7	88.5	91.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.1	10.7	9.4	69.0	98.1	1.9	2.3	67.6	99.8	66.5	101.1	65.5	102.4	64.1	104.1		
(5)			67.5	82.8	2.3	1.9	65.9	84.1	64.6	85.3	63.3	86.3	61.7	87.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	82.1	80.8	80.6	80.7	81.0	81.1	81.5	82.2	83.7	84.3	84.1	83.1	81.6	
	DBStd	2.80	2.47	2.64	2.41	2.21	2.05	2.18	2.22	2.30	2.70	2.89	2.73	2.65	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDD50	11705	955	857	951	930	964	945	997	1045	1030	1057	994	980	
	CDD65	6230	490	437	486	480	499	495	532	580	580	592	544	515	
	CDH74	64079	4358	3810	4256	4363	4752	5031	5725	6805	7006	6921	6027	5025	
	CDH80	22998	1268	1084	1162	1209	1341	1618	2075	2896	3183	3070	2446	1646	
Wind	WSAvg	3.9	3.9	4.1	4.1	3.6	3.4	3.6	3.8	4.1	4.2	4.3	4.1	3.9	
Precipitation	PrecAvg	90.1	11.2	11.2	12.8	12.6	9.6	4.6	3.0	2.2	2.9	4.5	7.2	9.4	
	PrecMax	124.3	22.4	24.4	24.9	21.1	21.9	9.0	8.1	5.6	9.6	10.0	13.3	20.4	
	PrecMin	69.4	1.3	4.3	5.3	5.4	2.6	0.4	0.1	0.1	0.0	0.4	0.5	3.5	
	PrecStd	11.5	3.5	4.3	4.3	3.9	3.9	1.9	1.9	1.6	2.0	2.4	3.4	4.0	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	92.2	92.4	91.7	92.3	91.1	91.3	93.0	94.9	97.0	96.7	95.2	93.4	
		MCWB	78.6	79.4	79.8	80.2	79.7	78.8	78.2	79.0	77.4	77.8	78.3	78.5	
	2%	DB	89.8	89.8	89.7	89.5	89.3	89.7	91.1	93.2	94.9	94.8	93.3	91.4	
		MCWB	78.3	78.2	78.9	79.1	79.0	78.3	77.8	78.1	78.0	78.5	78.5	78.3	
	5%	DB	88.1	88.0	87.8	87.9	87.8	88.2	89.7	91.7	93.2	93.2	91.6	89.5	
		MCWB	78.1	78.1	78.6	78.8	78.8	78.2	77.5	77.8	78.2	78.4	78.5	78.4	
	10%	DB	86.2	86.1	86.0	86.1	86.2	87.5	88.1	90.2	91.5	91.4	89.8	87.7	
		MCWB	77.9	77.8	78.1	78.4	78.5	78.1	77.5	77.7	78.1	78.3	78.4	78.3	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.8	81.2	81.1	81.6	80.9	80.5	80.1	80.8	81.0	81.4	81.0	81.0
			MCDB	87.8	88.3	88.4	88.5	87.7	87.7	88.2	90.0	90.1	90.3	89.7	88.2
2%		WB	79.6	79.8	79.9	80.1	79.9	79.6	79.2	79.8	80.1	80.4	80.2	79.9	
		MCDB	86.2	86.3	86.6	86.7	86.2	86.6	87.2	89.1	89.4	89.7	88.6	86.9	
5%		WB	78.9	79.0	79.0	79.2	79.2	79.0	78.8	79.1	79.5	79.7	79.7	79.2	
		MCDB	85.3	85.0	85.3	85.2	85.2	85.9	86.8	88.2	88.6	88.8	87.8	85.9	
10%		WB	78.3	78.3	78.4	78.7	78.8	78.5	78.2	78.6	78.9	79.1	79.1	78.7	
		MCDB	84.2	83.9	84.0	84.3	84.6	85.1	85.8	87.4	88.0	87.9	86.9	85.2	
Mean Daily Temperature Range	5% DB	MDBR	11.0	10.9	10.8	10.9	10.5	11.3	12.3	13.7	14.2	14.0	13.0	11.6	
		MCDBR	13.4	13.3	13.4	13.1	12.4	12.5	13.7	14.8	15.8	15.8	14.9	14.0	
		MCWBR	5.1	5.1	5.3	5.3	4.9	4.9	5.1	5.2	5.6	5.3	5.1	5.3	
	5% WB	MCDBR	11.8	11.9	12.2	11.8	11.4	11.9	12.8	14.0	14.6	14.7	13.7	12.5	
		MCWBR	5.2	5.3	5.3	5.3	5.0	5.1	5.2	5.4	5.8	5.8	5.6	5.4	
Clear-Sky Solar Irradiance	taub	0.399	0.393	0.393	0.385	0.387	0.390	0.403	0.488	0.529	0.508	0.494	0.452		
	taud	2.520	2.546	2.553	2.585	2.568	2.560	2.498	2.221	2.089	2.170	2.207	2.348		
	Ebn at Noon	297	298	295	290	281	276	274	257	254	263	267	279		
	Edh at Noon	35	35	34	32	31	31	33	45	53	50	48	42		
All-Sky Solar Radiation	RadAvg	1366	1353	1359	1336	1316	1460	1568	1682	1682	1605	1483	1331		
	RadStd	113	106	101	51	84	89	77	95	127	86	98	112		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	-0.40	-0.20	-2.93	-0.94	-2.07	-2.27	N/A	N/A	-275	-275

Nomenclature: See separate page